Extending DREM to Methylation Dynamics: A TF–DMR Prior Framework for Brain Cell Lineages

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Abstract

The Dynamic Regulatory Events Miner (DREM) is an Input-Output Hidden Markov Model originally formulated over time-series gene expression, in which a per-time-point dynamic measurement is paired with a static TF-to-gene prior. This report describes a methodological extension of the DREM framework to a different dynamic substrate: DNA methylation β-values measured along the developmental trajectories of human brain cell lineages. Because DREM was designed assuming expression as the dynamic input, this work also evaluates whether DREM remains functional when methylation, rather than expression, is supplied as the temporal substrate. The substrate change forces a corresponding change in the prior: the TF-to-gene table is replaced by a TF-to-DMR table built by intersecting ChIP-Atlas peak BEDs with lineage-specific DMR BEDs, retaining DMR identity rather than collapsing to gene-level annotations. This report describes the TF–DMR prior framework, the curated accession whitelist (172 records, including both SRX and ERX entries) that defines the transcription factor coverage, a BrainSpan RPKM expression filter that restricts the prior to TFs operative in developing human brain, and a parameterized regeneration pipeline that produces five lineage-specific TF–DMR priors used in a BICAN methylation analysis. Applied to neural differentiation trajectories in the BICAN human brain methylation atlas, the resulting DREM models recovered polycomb-antagonizing SWI/SNF chromatin-remodeler subunits (SMARCA4, SMARCB1) and early patterning TFs (ASCL1, SOX2) at regions gaining mCG in mid-gestation, identified SOX10 and MEF2C as lineage-specific drivers of methylation change in the OPC-to-ODC and MGE-derived inhibitory neuron arcs respectively.

1 Introduction

Dynamic regulatory networks underlie the transitions that distinguish one neural cell type from another during human brain development. A standard inferential question is which transcription factors (TFs) drive the bifurcations along a lineage trajectory, given a dynamic measurement of cell state and a candidate set of TF binding events. DREM [3, 10] addresses this question for time-series gene expression: an Input-Output Hidden Markov Model fits the temporal expression of each gene, and a logistic regression at each split node uses TF–gene binding evidence to annotate the bifurcation. The framework is agnostic about which biological measurement enters the dynamic input layer, but the default workflow assumes the dynamic substrate is gene expression and the prior connects TFs to genes through promoter windows.

The BICAN (Brain Initiative Cell Atlas Network) U01 multimodal atlas of the developing human brain [13] produces a different dynamic substrate: DNA methylation β-values measured at differentially methylated regions (DMRs) along each cell-type lineage. Methylation changes are localized to the DMRs themselves, not to the ± 2 kb windows around transcription start sites that the default TF–gene prior is built from. Applying DREM to methylation dynamics therefore requires replacing the prior. The substantive question is not how to fit the IOHMM but how to construct a prior whose substrate matches the substrate of the dynamic measurement.

2 Background

2.1 DREM and the IOHMM framework

The Dynamic Regulatory Events Miner (DREM) was introduced by Ernst et al. [3] as a method for reconstructing dynamic gene regulatory networks (GRNs) from time-series expression data. The model is an Input-Output Hidden Markov Model (IOHMM) over the temporal expression profile of each gene. Hidden states correspond to nodes in the dynamic network; observations are the per-time-point expression values of a gene; and the transition probabilities at each node are produced by a logistic regression classifier whose inputs are protein-DNA interaction values (the static input layer). When a node has more than one outgoing transition, that node is a *split node*: a point in pseudotime at which a co-expressed group of genes diverges into multiple paths. The TF annotations at each split are based on the assignment of genes to their most likely path, which is determined by inference over the entire model. The logistic regression at each split then identifies which transcription factors (TFs) most distinguish genes assigned to one outgoing path from genes assigned to the other [3, 10].

DREM 2.0 [10] extended the original framework with packaged protein-DNA interaction data for several species, support for continuous binding values, an option to scale TF effects by their own expression at each time point, dynamic (time-point-specific) input layers, and integrated motif discovery. The interactive GUI for inspection of the learned model was present in the original DREM and received smaller modifications in 2.0. The two pieces of input that the user supplies for a typical run are (i) a time-series matrix of dynamic measurements indexed by gene and time point, and (ii) a static or dynamic prior file in three columns: `TF`, `Gene`, `Input`, where the `Input` column is the binding value used by the logistic regression classifier (binary or continuous).

The first input encodes *what is changing* along the trajectory. The second input encodes *which regulators are candidates to explain the change*. DREM was originally designed assuming the first input is gene expression; whether it remains functional when methylation is supplied as the dynamic substrate instead is one of the questions this work evaluates. In principle the framework is agnostic about what biological measurement appears in the first input as long as it varies along time or pseudotime, and about what the prior connects as long as the rows of the dynamic matrix line up with the entities the prior maps to. Although DREM 2.0 supports dynamic (time-point-specific) priors, the work here uses a single static prior throughout. This separation between the dynamic substrate and the static prior, together with a correspondence between the row identifiers of the dynamic matrix and the entities indexed in the prior, is what the modality extension exploits.

2.2 Methylation dynamics in the BICAN brain atlas

The BICAN U01 project assembled a multi-modal single-cell atlas of the developing human brain spanning mid-gestation through 7 months postnatal, profiling DNA methylation (mCG) and three-dimensional chromatin contacts (3C) jointly via snm3C-seq [13]. The developmental BICAN data were integrated with previously collected adult brain data, so that each lineage spans from a fetal progenitor state through to the corresponding adult cell type. Within that atlas, lineage trajectories were constructed for major neural cell types, including medial ganglionic eminence (MGE)-derived parvalbumin (PV) interneurons (MGE_PVALB), two medium spiny neuron (MSN) subtypes (MSN_DRD1_BACH2 and MSN_DRD1_EPHA4), the oligodendrocyte precursor cell (OPC) to oligodendrocyte (ODC) differentiation arc (OPC_ODC), and an embryonic caudal ganglionic eminence (eCGE)-derived LAMP5 inhibitory neuron lineage (eCGE_LAMP5).

Differentially methylated regions (DMRs) were called by methylpy pairwise between adjacent developmental stages along each lineage (3T → 1m → 4–7m → adult). For each pairwise comparison, methylpy reports the differential regions from the perspective of each side, labeling both directions as “hypo” relative to the higher-methylation side. The merged lineage BED used here (despite its `_hypo_merged` filename) is the union of both directions of those pairwise comparisons, so it captures regulatory elements that lose mCG and regulatory elements that gain mCG along the trajectory, rather than demethylation events alone.

The dynamic measurement of interest in this setting is methylation β at each DMR, indexed by developmental-stage cluster ordered along the lineage trajectory. This is the natural input to a DREM-style IOHMM if one wants to ask which TFs drive the methylation transitions at lineage bifurcations. The challenge is that DREM’s standard prior is a TF-to-gene table built by intersecting ChIP-seq peaks with promoter windows, which is appropriate when the dynamic substrate is gene expression but mismatched when the dynamic substrate is methylation: the regions where methylation is changing are the DMRs themselves, not the ±2 kb windows around transcription start sites. Resolving this mismatch is the substantive methodological problem the rest of this report addresses.

2.3 ChIP-Atlas and the TF binding evidence base

The TF binding evidence used to build the prior comes from ChIP-Atlas [8], a public reanalysis pipeline that uniformly processes ChIP-seq, ATAC-seq, and DNase-seq experiments deposited in NCBI SRA and similar archives. For each experiment, ChIP-Atlas publishes a peak BED file at multiple stringency thresholds and an experiment metadata table linking each experiment accession (SRX, ERX, or DRX, depending on the archive in which the experiment was originally deposited) to a target antigen and cell type. The peak BEDs are the substrate from which the TF-DMR prior is constructed: the framework intersects per-experiment peaks with lineage-specific DMR BEDs to produce a (TF, DMR) graph weighted by which transcription factors show binding evidence in the genomic regions where methylation is dynamic along each lineage. ChIP-Atlas’s metadata also requires curation, since the “antigen” column includes entries that are not sequence-specific transcription factors (epitope tags, oxidative modifications, GFP fusions, histone modifications); these typically reflect a different experimental approach in which the intended target is annotated in a different metadata column, rather than annotation errors. The accession set used here filters these out; the antigens that remain are predominantly sequence-specific transcription factors, together with a minority of non-sequence-specific chromatin-associated proteins (for example cohesin and Mediator subunits and SWI/SNF remodeler subunits) that the curation procedure retained on the strength of their ChIP-seq evidence. Section 3.4 enumerates this composition.

2.4 BrainSpan as a brain-relevance filter

ChIP-Atlas aggregates experiments across all available cell types and tissues, so a TF that has many peaks in the genome may nonetheless be biologically irrelevant in the brain. To restrict the prior to TFs that are operative in human neural tissue, this work uses the BrainSpan developmental transcriptome [7] as a second filter. BrainSpan provides bulk RNA-seq from 524 samples spanning multiple brain regions and developmental stages from 8 post-conceptional weeks through adulthood, processed to a per-gene RPKM matrix. A TF passes the filter if its mean RPKM across the BrainSpan samples exceeds a threshold. The threshold is chosen so that borderline regulators of demonstrated brain importance are retained: REST, for example, has a BrainSpan mean RPKM of 1.29, and the threshold used here is set just below such biologically anchored points so that REST and similar regulators are not excluded. The choice is discussed in Section 3.

2.5 Related approaches

Several approaches use temporal expression data to reconstruct regulatory networks. Friedman’s dynamic Bayesian networks [4] provide a formally clean framework for learning network structure from time-series expression but do not natively incorporate a static TF-DNA prior, and the structure-learning scales poorly to the hundreds of TFs that typical regulatory analyses require. Network Component Analysis [5] does combine static interaction data, in the form of a TF-promoter connectivity matrix derived from ChIP-chip or literature, with expression measurements, but its matrix-factorization decomposition treats the time index as exchangeable rather than ordered. DREM’s IOHMM occupies a useful middle ground: it respects time ordering, scales to hundreds of TFs through the logistic regression at split nodes, and produces an interpretable graphical model with TF annotations attached to specific bifurcation events. The work presented here treats DREM’s IOHMM as a fixed inferential engine and focuses entirely on the prior side of the input pair, swapping the gene-substrate prior for a DMR-substrate prior matched to the methylation dynamic measurement.

3 The TF–DMR Prior Framework

3.1 DMR β-matrix construction

The dynamic input layer that DREM observes for each lineage is a matrix of methylation β-values indexed by DMR (rows) and developmental-stage cluster (columns). Construction of this matrix begins upstream of the prior, from per-cluster allc-format methylation files produced by the BICAN snm3C-seq pipeline. Each allc file records, for every cytosine position called in that cluster, the chromosome, position, strand, sequence context, the count of methylated reads m_C, and the total read count C at that position. The construction pipeline operates per lineage; its inputs are (i) a lineage cluster-order TSV listing the per-cluster allc files in order along the developmental trajectory, and (ii) the merged lineage DMR BED. The merged BED is the union of the pairwise DMRs called between adjacent developmental stages (Section 2), so its rows include regions that lose mCG and regions that gain mCG along the lineage.

(1) For each developmental-stage cluster file, extract the calls in CpG sequence context (allc encodes these with a context prefix `CG...`) and reformat them into BED-like rows `chr start end` m_C C, sorted by genomic coordinate.
(2) Aggregate the per-cluster CpG file over the merged lineage DMR BED with `bedtools map -sorted`, computing for each DMR the sum of methylated reads $\sum m_C$, the sum of total reads $\sum C$, and the count of overlapping CpGs.
(3) Compute the per-DMR, per-cluster β value as the coverage-weighted ratio

$$\beta_{d,c} = \frac{\sum_{i\in d\cap c} m_{C,i}}{\sum_{i\in d\cap c} C_i},$$

where the sum runs over CpG sites i that lie within DMR d and have coverage in cluster c. A DMR with no covered CpGs in a given cluster has no defined β for that cell. DREM requires a missing value to be an empty field rather than a literal placeholder string; rather than rely on that convention, any DMR whose row would contain a missing value in one or more clusters is removed as an intermediate filtering step before the matrix is passed to DREM, so the matrix DREM observes carries a defined β in every cell.
(4) Assemble the DMR-by-cluster β matrix by pasting the per-cluster β columns side-by-side in order along the developmental trajectory, using the cluster-order TSV as the column ordering.
(5) Restrict the assembled matrix to the 50,000 DMRs with the largest max-minus-min change in β across the developmental-stage clusters. The full per-lineage DMR set is large (roughly 0.5 to 1.6 million DMRs depending on lineage), so this step selects the most dynamic DMRs; the resulting 50,000-row matrix is the DREM observation layer for that lineage.

The aggregation in step 3 is coverage-weighted at the DMR level rather than computed by averaging per-CpG β values. A CpG that contributes 30 reads to the DMR and a CpG that contributes 3 reads enter the DMR-level β in proportion to their coverage; this matches how β is defined per CpG and avoids the bias that uniform-weight averaging across CpGs of widely varying coverage would introduce. The same pipeline runs once per lineage with the matching cluster-order TSV (e.g., `bican_OPC_ODC.order.tsv`) and merged lineage DMR BED; per-lineage scripts `apm_<lineage>_DMR.sh` and `combine_filtered_methylation_<lineage>.sh` parameterize the per-lineage inputs (the cluster-order TSV listing the per-cluster allc files in developmental-stage order, and the lineage DMR BED), the lineage labels, and the output paths across the five lineages used in this analysis. The core aggregation step is given as pseudocode in Section 4 (Algorithm 1).

3.2 Matching the prior substrate to the dynamic substrate

DREM's static input layer connects regulators to the entities whose dynamic values appear in the IOHMM observation layer. In the standard expression workflow, the observation layer carries per-time-point expression values for each gene, and the prior connects each TF to the genes for which it has ChIP-seq peak evidence in the ±2 kb promoter window. The substrate of the prior matches the substrate of the dynamic measurement: both are indexed by gene.

When the dynamic measurement is methylation β at lineage DMRs along the developmental trajectory, the substrate of the dynamic measurement is the DMR set, not the gene set. The regions where methylation is actually changing across the trajectory are the DMRs; gene-promoter windows are at best a coarse proxy and at worst a different biological signal. A prior that connects TFs to genes is therefore mismatched: it asks the IOHMM split classifier to learn associations between regulators and entities (genes) that are not the entities being measured (DMRs). The TF–DMR prior framework resolves the mismatch by building the prior over the same substrate as the dynamic measurement: ChIP-Atlas peaks are intersected directly with the merged lineage DMR BED, and the resulting prior maps each `TF|cell|accession` key to the set of DMR identifiers where that experiment's peaks overlap a lineage DMR.

3.3 Construction pipeline

The construction pipeline operates per lineage. The inputs are (i) a merged lineage DMR BED file, (ii) a list of ChIP-Atlas top-1000 peak BED paths, and (iii) a curated accession-to-(TF, cell) label table. The steps are:

(1) Sort the merged lineage DMR BED by genomic coordinate (`bedtools sort`).
(2) For each whitelisted accession, intersect the ChIP-Atlas peak BED with the sorted DMR set using `bedtools intersect -wa -wb`, retaining the accession alongside the overlapping DMR identifier.
(3) Deduplicate the resulting (accession, `dmr_id`) pairs.
(4) Join the (accession, `dmr_id`) pairs to the curated accession-to-(TF, cell) label table on accession.
(5) Emit two outputs in DREM 3-column format: an annotated prior keyed by `TF|cell|accession` and a collapsed prior keyed by TF only. Only the annotated prior is used in the analysis (Section 3.5).

Figure 1 summarizes the pipeline. The core intersection step is given as pseudocode in Section 4 (Algorithm 2).

3.4 Curated TF set

The curated TF set defines which ChIP-Atlas experiments enter the TF–DMR prior. The curation procedure proceeds as follows. For each (TF, cell type) combination that passes the curation criteria, the three ChIP-Atlas experiments with the highest average binding enrichment are retained; this top-3 selection rule matches the per-state TF assignment used in the ChromHMM-based [2] BICAN methylation analysis [13]. Taking the union of these retained experiments across all (TF, cell type) combinations yields the 172-record accession whitelist used as input to the TF–DMR pipeline. Because ChIP-Atlas aggregates experiments deposited in several short-read archives, the curated accessions are not all SRX-prefixed: they include SRX and ERX (and similar) accessions, reflecting the archive in which each experiment was originally deposited. The non-SRX accessions were part of the original selection procedure rather than a separate addendum. In total the curated set comprises 172 accessions, including both SRX and ERX entries.

The exclusion criteria remove categories of antigens that appear in the ChIP-Atlas metadata but are not informative for TF–DMR inference: antibody-tag entries (`Epitope`, `GFP`, `8-Hydroxydeoxyguanosine` and similar) that reflect different experimental approaches where the TF target was annotated in a different ChIP-Atlas metadata column, and histone-modification antigens (`H3K27ac`, `H3K4me3`, `H3K9me3` and similar), which describe chromatin state rather than sequence-specific binding. Inspection of the final curated set confirms that no histone-modification antigen remains in it.

The retained antigens are not exclusively sequence-specific transcription factors. Most are sequence-specific TFs (for example REST, SOX10, SOX2, ASCL1, NEUROD1, FOS, FOSL2, JUNB, the MEF2 family, and nuclear receptors such as NR3C1). The remainder are non-sequence-specific chromatin-associated proteins that the top-3 average-binding-enrichment procedure retained because they carry strong, localized ChIP-seq evidence: the cohesin subunits RAD21 and SMC1A, which form a structural, architectural complex rather than a transcriptional cofactor; the Mediator subunits MED1 and MED12, which form a transcriptional coactivator complex; ATP-dependent SWI/SNF–BAF remodeler subunits (SMARCA4, SMARCB1, SS18, BRD9); chromatin readers (BRD4, CBX2); chromatin- and DNA-modifying enzymes (EP300, DNMT3A, SETD1A); and transcription-elongation and RNA-associated factors (NELFE, SETX, FUS, EWSR1). These are retained deliberately, as proteins whose localized binding is informative about regulatory-element activity at DMRs, but they are not sequence-specific regulators and are described as such rather than as "cofactors." One entry, `BCL10`, is a cytoplasmic signaling adaptor with no transcriptional or chromatin-associated role; it is a curation artifact rather than a regulatory input and should be dropped from the curated set. The curation is canonical across all five lineage outputs: the same accession-to-(TF, cell) mapping is applied to every lineage, and the final TF–DMR files exhibit zero label conflicts across the five outputs.

3.5 BrainSpan expression filter

ChIP-Atlas aggregates experiments across all available cell types and tissues, so a TF can pass the curation step on the strength of peaks in a non-brain context. The BrainSpan filter restricts the prior to TFs that are expressed in developing human brain. The filter rule is:

(1) Compute, for each row of the BrainSpan expression matrix, the mean RPKM across all 524 BrainSpan samples.
(2) For each gene symbol with multiple transcript-level rows, take the maximum mean RPKM across rows. A TF counts as expressed if any of its transcript rows meets the threshold.
(3) Retain a TF if its per-symbol mean RPKM exceeds the threshold.

The threshold used is mean RPKM > 1.0. Its rationale is the borderline-regulator anchor point: REST, a transcriptional repressor with well-characterized roles in neural lineage specification, has a BrainSpan mean RPKM of 1.29, so a threshold above that value would drop REST despite its biological relevance, while 1.0 retains REST and similar borderline regulators yet still excludes TFs that are not appreciably expressed across the BrainSpan samples. Of the 78 unique TF symbols in the curated gene-substrate prior used as the analog reference, 58 pass the filter, 17 fall below the threshold, and 3 are absent from the BrainSpan gene table entirely: the antibody-tag antigens `Epitope` and `GFP`, which have no gene-symbol counterpart, and `KMT2D`, a gene symbol not represented in the BrainSpan rows metadata. The analysis in this report uses a single prior variant: the annotated `TF|cell|accession` prior, which is the file supplied to DREM as its TF-interaction input. The BrainSpan filter operates on this prior by extracting the TF symbol as the substring preceding the first `|` in the first column. The regeneration script also emits two further products that were *not* used in the DREM analysis: a TF-collapsed prior keyed by TF symbol only, and a protein-coding-restricted variant of the annotated prior. The protein-coding-restricted variant is kept only as a robustness check; applying the same filter to it yields the same retained set, because the protein-coding restriction is a strict subset of the BrainSpan filter on the TF column.

4 Implementation

The pipeline is implemented as bash plus Python, with all genomic operations delegated to `bedtools` [9] and the BrainSpan filter implemented in `pandas` [6]. The full pipeline runs on UCLA's Hoffman2 cluster; per-lineage runtime is dominated by the `bedtools intersect` step over the ChIP-Atlas peak set, which parallelizes naturally over accessions. The scripts that implement each step are maintained in the project repository; rather than reproduce the shell plumbing, this section gives pseudocode for the two steps where the algorithm itself is the point.

Algorithm 1 gives the per-cluster CpG aggregation over DMRs that produces one column of the β-matrix described in Section 3.1. For each developmental-stage cluster, CpGs are extracted from the allc file and mapped over the sorted merged lineage DMR BED, the methylated and total reads are summed and overlapping CpGs counted per DMR, and the per-DMR sums are converted into a coverage-weighted β value.

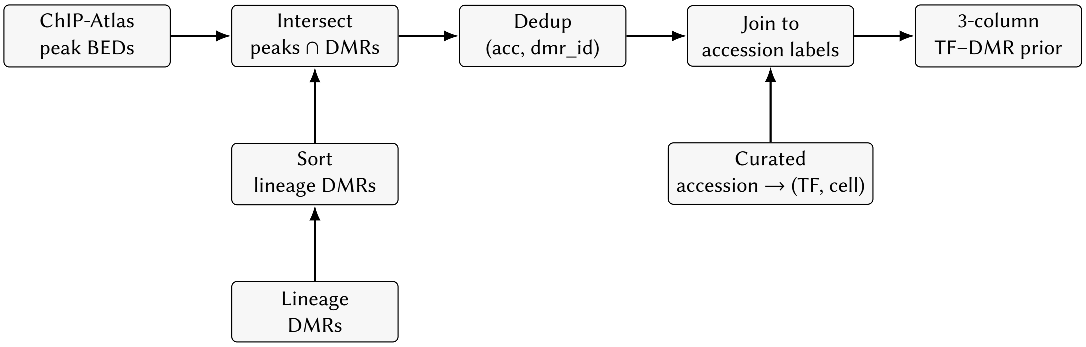

Figure 1: TF–DMR prior construction pipeline. ChIP-Atlas peak BEDs feed directly into the intersection and are shared across lineages; the lineage DMRs are sorted on their own path before the intersection. The lineage DMR BED and the curated accession label table are the lineage-specific and curation-specific inputs respectively.

Algorithm 1 Per-cluster coverage-weighted β over the merged lineage DMR BED (implemented in `apm_<lineage>_DMR.sh`).

Require: sorted merged lineage DMR BED D; sorted per-cluster CpG records, each overlapping CpG i carrying methylated reads m_i and total reads C_i
Ensure: per-DMR β for this cluster

for each DMR $d \in D$ **do**
  $M_d \leftarrow \sum\{\, m_i : \text{CpG } i \text{ overlaps } d \,\}$
  $T_d \leftarrow \sum\{\, C_i : \text{CpG } i \text{ overlaps } d \,\}$
  $n_d \leftarrow |\{\, i : \text{CpG } i \text{ overlaps } d \,\}|$
  if $T_d = 0$ **or** $n_d = 0$ **then**
    $\beta_d \leftarrow \text{NA}$
  else
    $\beta_d \leftarrow M_d / T_d$
  end if
end for
return $\{(d, \beta_d) : d \in D\}$

Algorithm 2 Peak-DMR intersection (implemented in `regenerate_tf_dmr.sh`).

Require: whitelisted accessions A; per-accession peak BED B_a; sorted merged lineage DMR BED D
Ensure: deduplicated set of (accession, DMR) pairs

$S \leftarrow \emptyset$
for each accession $a \in A$ **do**
  for each peak $p \in B_a$ overlapping some DMR $d \in D$ **do**
    $S \leftarrow S \cup \{(a,\ \text{id}(d))\}$
  end for
end for
return S

Algorithm 2 gives the core peak-DMR intersection. For each whitelisted accession, the peak BED is intersected with the sorted DMR set, the (accession, DMR identifier) pair is emitted for every overlap, and the cumulative output is deduplicated.

The BrainSpan filter follows the enumerated rule of Section 3.5: the expression matrix is averaged row-wise across all 524 samples, the row means are joined to the gene-symbol metadata, the maximum mean across transcript-level rows is taken per gene symbol, and TFs are retained where the per-symbol mean exceeds the threshold.

Listing 1 shows an excerpt of the OPC–ODC β-matrix produced by the pipeline of Section 3.1. Rows are DMRs and columns are developmental-stage clusters in lineage order; each cell is the coverage-weighted β for that DMR in that cluster. The matrix is the dynamic-input layer that DREM observes for the lineage.

Listing 2 shows the resulting 3-column TF–DMR prior file format. The first column carries the annotated `TF|cell|accession` key; the second column is the DMR identifier that the merged lineage DMR BED assigns to the overlapping region; the third column is the binary input value that DREM's logistic regression at the split nodes consumes. The accession in the excerpt below (`DRX190656`) is itself a non-SRX accession, illustrating that the curated set spans multiple short-read archives.

Listing 2: Excerpt of the regenerated MGE_PVALB TF–DMR prior file. Each row records that experiment `DRX190656` for TCF21 in HCF cells has a ChIP-Atlas peak overlapping the named DMR in the MGE_PVALB lineage.

```
TF                      Gene                          Input
TCF21|HCF|DRX190656     dmr_chr10_102208297_102209008    1
TCF21|HCF|DRX190656     dmr_chr10_103771954_103772499    1
TCF21|HCF|DRX190656     dmr_chr10_11242443_11242850      1
```

5 Results

The BrainSpan filter outcome on the analog gene-substrate reference (78 unique TF symbols; 58 retained, 17 below threshold, 3 absent from BrainSpan) is reported in Section 3.5; this section focuses on the downstream effect of the regenerated TF–DMR prior on the fitted DREM models.

The DREM model fit for the OPC–ODC trajectory using the regenerated TF–DMR prior is shown in Figure 2. The model spans seven developmental stages from second-trimester glial progenitor cells (2T_GPC) through progenitor and committed populations (GPC, OPC, tOPC, tODC) to mature and adult oligodendrocytes (ODC, adult_ODC), and resolves a primary bifurcation at the earliest stage transition that separates regulatory elements demethylating across the lineage (upper paths) from those gaining mCG (lower

Listing 1: Excerpt of the OPC–ODC DMR β-matrix (`OPC_ODC_dmr_beta.matrix.tsv`). Rows are DMRs; columns are developmental-stage clusters in lineage order; cells are coverage-weighted methylation β-values. Empty cells (omitted in this excerpt) indicate that no CpG within the DMR had coverage in the corresponding cluster.

```
                                2T_GPC  GPC     OPC     tOPC    tODC    ODC     adult_ODC
dmr_chr1_117587864_117588098    1       0.9545  0.0111  0.0172  0       0.0168  0.2190
dmr_chr10_96560195_96560440     1       0.8247  0.1667  0.0968  0       0.0563  0.1474
dmr_chr11_27669278_27669395     1       0.8333  0.1698  0.0769  0.1667  0       0.0588
```

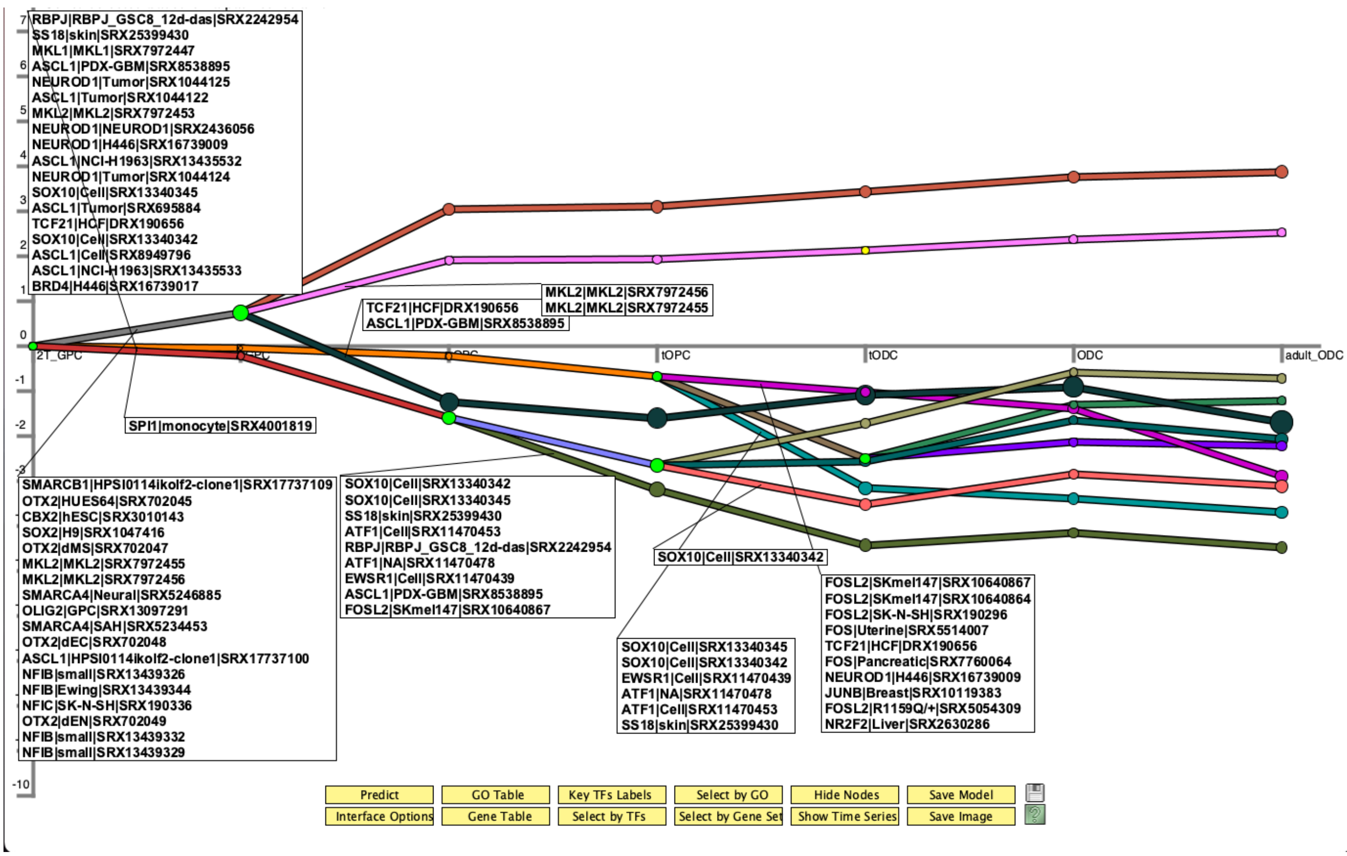

Figure 2: DREM model fit for the OPC–ODC trajectory using the regenerated TF–DMR prior. Developmental stages advance left-to-right across the lineage (2T_GPC, GPC, OPC, tOPC, tODC, ODC, adult_ODC). Split nodes are annotated with the transcription factors that the logistic regression at the split identifies as significant discriminators between genes assigned to the outgoing paths.

paths). OPC–ODC is the single lineage for which a DMR-substrate DREM model was fit; the other lineages have regenerated TF–DMR priors and β-matrices but no fitted DMR model. Quantitatively, this model is an IOHMM fit over 50,000 input DMRs, with 47 nodes in total: 6 split nodes (two two-way and four three-way), 11 terminal paths, and 10 logistic-regression discriminators (one for each outgoing branch beyond the first at a split, so that the per-split discriminator count is the split's out-degree minus one).

At the upper, demethylating split, the DREM model annotates the bifurcation with SOX10, NEUROD1, ASCL1, and TCF21 as significant regulators, drawing on the TF–DMR prior. A single ChIP-Atlas experiment is sufficient for SOX10 to be called at this split; here it is supported by multiple experiments, which is corroborative rather than required. SOX10 is also annotated independently at the OPC-to-tOPC and tOPC-to-tODC splits, indicating that the regenerated prior captures SOX10 binding enrichment at progressively demethylating regulatory elements across both OPC commitment and ODC maturation rather than only at the lineage-defining bifurcation. This identifies SOX10 as a lineage-specific driver of demethylation along the OPC–ODC trajectory, consistent with the role reported in the BICAN methylation manuscript [13] and with SOX10's established requirement for terminal oligodendrocyte differentiation and myelination [1, 12], and adds the observation that the signal is sustained across multiple intra-lineage transitions.

At the lower, methylation-gaining split, the model annotates the bifurcation with SMARCB1, SMARCA4, SOX2, OTX2, OLIG2, and NFIB. The SMARCA4 and SMARCB1 enrichments at regions gaining mCG in mid-gestational development are consistent with SWI/SNF chromatin-remodeler binding at sites where polycomb repression is progressively replaced by de novo methylation [11, 13]. The co-enrichment of SOX2 and ASCL1 at the same path is consistent with early patterning TF binding sites gaining mCG in mid-gestation [13]. OLIG2 and NFIB at this path are consistent with pro-glial factor binding sites accumulating methylation during cell-fate commitment [13], here visible from within the glial trajectory itself: regulatory elements bound by OLIG2 in glial progenitors silence as cells commit further along the OPC–ODC arc.

Taken together, the OPC–ODC DREM fit produced by the TF–DMR prior pipeline yields biologically coherent split annotations: a demethylating arc dominated by SOX10 and other glial-lineage regulators, and a methylation-gaining arc carrying SWI/SNF and early patterning factors. These are standalone findings of the pipeline rather than a reproduction of any prior model. The fit additionally resolves SOX10 enrichment across three successive intra-lineage splits. One caveat applies to all of these calls: the top-3 selection rule itself bounds TF coverage, so the annotations reflect only TFs with enough average binding enrichment in ChIP-Atlas to clear that cutoff; relaxing the top-3 restriction would admit additional candidate TFs at every split.

6 Discussion and Limitations

The TF–DMR prior framework is constrained by ChIP-Atlas coverage. A TF can only enter the prior if it has enough public ChIP-seq accessions to clear the top-3 selection criterion, regardless of its biological importance. TFs with no ChIP-seq evidence in ChIP-Atlas are absent from the prior even when there is independent evidence of their role in a lineage; this is a limitation inherited from the underlying TF binding evidence base, not from the prior construction itself. The top-3 rule is itself a restriction, not only a coverage floor: relaxing it (retaining more than three experiments per (TF, cell type) combination, or lowering the average-binding-enrichment bar) would admit additional TFs into every lineage prior, at the cost of including experiments with lower average binding enrichment.

The BrainSpan expression filter assumes that brain-wide bulk RPKM is a reasonable proxy for whether a TF is operative in any specific lineage. This is conservative in the right direction (a TF that is unexpressed across all 524 BrainSpan samples is unlikely to be operative in any lineage), but it does not distinguish between TFs that are broadly expressed and TFs that are sharply lineage-specific. A lineage-specific expression filter built from per-cell-type expression data would refine the prior further and is a natural next step.

The prior is binary at the TF level after collapse. The 3-column DREM input format admits continuous values in the third column, and DREM 2.0 [10] can in principle consume the ChIP-Atlas peak score as a continuous binding intensity. The current TF–DMR prior discards that information, treating an overlap as a binary indicator. Reintroducing the peak-strength signal is a mechanical change to the construction script but would require calibrating the input scale against the existing IOHMM logistic regression behavior.

DMR calling is performed upstream by the BICAN methylation pipeline and is not part of the framework presented here. The TF–DMR prior inherits whatever DMR-calling assumptions were used for the merged lineage DMR BEDs: developmental-stage binning, the pairwise between-stage comparison, coverage filters, methylation-difference thresholds, and the union-merging strategy. Changes to those upstream choices would propagate into a different DMR set and therefore a different prior, even with no change to the construction script.

7 Future Work

The same prior-construction infrastructure used here for methylation extends to scRNA-seq lineage trajectories. In that setting, the dynamic substrate is gene expression aggregated to the lineage trajectory clusters (the same developmental-stage / trajectory clustering used to order the methylation columns, not a separate pseudotime binning), and the prior reverts to a TF-to-gene table. The construction differs from the standard DREM workflow primarily in the accession curation and, where applicable, the BrainSpan filter. The accession curation transfers unchanged to any system; the BrainSpan filter transfers only when the new system is also a developing-brain context. For a non-brain atlas such as IGVF, BrainSpan is not an appropriate relevance filter and would be replaced by an expression reference matched to that system. That arm of the work is in progress with collaborator-provided trajectory data and will be reported separately in an in-progress writeup.

8 Summary

DREM's IOHMM is agnostic about the substrate of its dynamic input as long as the prior establishes a correspondence between transcription factors and the entities indexed in the dynamic matrix. The TF–DMR prior framework realizes that observation for the BICAN methylation analysis: it replaces the standard TF-to-gene prior with a TF-to-DMR prior, restricts the TF coverage with a curated accession whitelist (172 records, including both SRX and ERX entries) and a BrainSpan RPKM filter, and produces all five lineage-specific priors from a single parameterized regeneration pipeline. The framework also provides a working test of whether DREM remains functional with methylation, rather than expression, as the dynamic substrate. The regenerated priors retain the biologically anchored borderline regulators that motivated the threshold choice and yield an OPC–ODC DREM fit with biologically coherent split annotations, including SOX10 enrichment resolved across multiple intra-lineage splits.

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References

[1] Ben Emery and Q. Richard Lu. 2015. Transcriptional and Epigenetic Regulation of Oligodendrocyte Development and Myelination in the Central Nervous System. *Cold Spring Harbor Perspectives in Biology* 7, 9 (2015), a020461. doi:10.1101/cshperspect.a020461

[2] Jason Ernst and Manolis Kellis. 2017. Chromatin-state discovery and genome annotation with ChromHMM. *Nature Protocols* 12, 12 (2017), 2478–2492. doi:10.1038/nprot.2017.124

[3] Jason Ernst, Oded Vainas, Christopher T. Harbison, Itamar Simon, and Ziv Bar-Joseph. 2007. Reconstructing dynamic regulatory maps. *Molecular Systems Biology* 3 (2007), 74. doi:10.1038/msb4100115

[4] Nir Friedman, Kevin Murphy, and Stuart Russell. 1998. Learning the structure of dynamic probabilistic networks. (1998), 139–147.

[5] James C. Liao, Riccardo Boscolo, Young-Lyeol Yang, Linh M. Tran, Chiara Sabatti, and Vwani P. Roychowdhury. 2003. Network component analysis: Reconstruction of regulatory signals in biological systems. *Proceedings of the National Academy of Sciences* 100, 26 (2003), 15522–15527. doi:10.1073/pnas.2136632100

[6] Wes McKinney. 2010. Data Structures for Statistical Computing in Python. In *Proceedings of the 9th Python in Science Conference (SciPy)*. 56–61. doi:10.25080/Majora-92bf1922-00a

[7] Jeremy A. Miller, Song-Lin Ding, Susan M. Sunkin, Kimberly A. Smith, Lydia Ng, Aaron Szafer, Amanda Ebbert, Zackery L. Riley, Joshua J. Royall, Kaylynn Aiona, et al. 2014. Transcriptional landscape of the prenatal human brain. *Nature* 508, 7495 (2014), 199–206. doi:10.1038/nature13185

[8] Shinya Oki, Tazro Ohta, Go Shioi, Hideki Hatanaka, Osamu Ogasawara, Yoshihiro Okuda, Hideya Kawaji, Ryo Nakaki, Jun Sese, and Chikara Meno. 2018. ChIP-Atlas: a data-mining suite powered by full integration of public ChIP-seq data. *EMBO Reports* 19, 12 (2018), e46255. doi:10.15252/embr.201846255

[9] Aaron R. Quinlan and Ira M. Hall. 2010. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* 26, 6 (2010), 841–842. doi:10.1093/bioinformatics/btq033

[10] Marcel H. Schulz, William E. Devanny, Anthony Gitter, Shan Zhong, Jason Ernst, and Ziv Bar-Joseph. 2012. DREM 2.0: Improved reconstruction of dynamic regulatory networks from time-series expression data. *BMC Systems Biology* 6 (2012), 104. doi:10.1186/1752-0509-6-104

[11] Benjamin Z. Stanton, Courtney Hodges, Joseph P. Calarco, Simon M. G. Braun, Wai Lim Ku, Cigall Kadoch, Keji Zhao, and Gerald R. Crabtree. 2017. Smarca4 ATPase mutations disrupt direct eviction of PRC1 from chromatin. *Nature Genetics* 49, 2 (2017), 282–288. doi:10.1038/ng.3735

[12] C. Claus Stolt, Silke Rehberg, Marius Ader, Petra Lommes, Dieter Riethmacher, Melitta Schachner, Udo Bartsch, and Michael Wegner. 2002. Terminal differentiation of myelin-forming oligodendrocytes depends on the transcription factor Sox10. *Genes & Development* 16, 2 (2002), 165–170. doi:10.1101/gad.215802

[13] The BICAN U01 Consortium. 2026. Multi-modal single-cell epigenomic atlas of the developing human brain. Preprint. Manuscript in preparation, 2026. TODO: add preprint venue and DOI/URL.